Unit 1 Homework — Real Numbers — Answer Key

Complete independently · Converting · Classification · Squares · Estimating Roots · Ordering

A. Converting Fractions ↔ Decimals

Problem 1 Score 3

Convert $\frac{7}{8}$ to a decimal.

Answer: 0.875

Problem 2 Score 3

Convert 0.125 to a fraction in simplest form.

Answer: $\frac{1}{8}$

Problem 3 Score 4

Convert $\frac{5}{9}$ to a decimal.

Answer: $0.\overline{5}$

B. Classifying Numbers

Problem 1 Score 3

Is √20 rational or irrational?

Answer: irrational

Problem 2 Score 3

Is √49 rational or irrational?

Answer: rational

Problem 3 Score 4

Is $-\frac{3}{4}$ rational or irrational?

Answer: rational

Problem 4 Score 4

Is 0.1010010001... (non-repeating pattern) rational or irrational?

Answer: irrational

Problem 5 Score 4

Name the smallest set of numbers that −8 belongs to: natural, whole, integer, or rational?

Answer: integer

C. Perfect Squares

Problem 1 Score 3

What is 15^2?

Answer: 225

Problem 2 Score 3

What is √196?

Answer: 14

D. Estimating Square Roots

Problem 1 Score 3

Between which two consecutive integers does √40 fall?

Answer: 6 and 7

Problem 2 Score 4

Estimate $\sqrt{40}$ to the nearest tenth.

Answer: 6.3

E. Ordering Rational & Irrational Numbers

Problem 1 Score 4

Which is greater: $\sqrt{5}$ or 2.2?

Answer: $\sqrt{5}$

Problem 2 Score 5

Order from least to greatest: $\frac{7}{2}$, $\sqrt{11}$, 3.4

Answer: $\sqrt{11}$, 3.4, $\frac{7}{2}$

Problem 3 Score 5

Order from least to greatest: $-\sqrt{4}$, $-\frac{3}{2}$, -1

Answer: $-\sqrt{4}$, $-\frac{3}{2}$, -1